

Full Stack Task Manager Project Report

This report explains the development of a Full Stack Task Manager application using modern web technologies. The application allows users to create, update, delete, and manage daily tasks efficiently. The project demonstrates frontend development, backend APIs, database integration, and deployment workflow.

1. Project Overview

The Task Manager application is a productivity tool that helps users organize tasks. Users can add tasks, set priorities, update status, and delete completed tasks. The application follows Full Stack architecture with frontend, backend, and database layers.

2. Technologies Used

Frontend: React.js, HTML5, CSS3, JavaScript
Backend: Node.js and Express.js
Database: MongoDB
Authentication: JWT (JSON Web Token)
Version Control: Git and GitHub
Deployment: Vercel (Frontend) and Render/Heroku (Backend)

3. System Architecture

The frontend communicates with the backend through REST APIs. The backend processes requests and stores task data in MongoDB. Authentication middleware secures protected routes.

4. Features of the Task Manager

- User Registration and Login
- Create New Tasks
- Update Task Details
- Delete Tasks
- Task Completion Tracking
- Responsive User Interface
- Secure Authentication

5. Frontend Development

React.js is used to build reusable UI components. Axios is used for API communication. React Router enables page navigation. State management is handled using React Hooks.

6. Backend Development

Node.js with Express.js handles server-side logic. REST APIs are created for CRUD operations. Middleware handles authentication and validation. Error handling improves application stability.

7. Database Design

MongoDB stores user and task information. Collections: • Users Collection • Tasks Collection

8. API Endpoints

POST /register – Register user

POST /login – User login

GET /tasks – Fetch tasks

POST /tasks – Add task

PUT /tasks/:id – Update task

DELETE /tasks/:id – Delete task

9. Sample Frontend Code

```
const addTask = async () => {  
  await axios.post('/tasks', {  
    title: taskTitle,  
    completed: false  
  });  
};
```

10. Sample Backend Code

```
app.post('/tasks', async (req, res) => {  
  const task = new Task(req.body);  
  await task.save();  
  res.send(task);  
});
```

11. Advantages of the Project

- Improves productivity
- Easy task organization
- User-friendly interface
- Scalable architecture
- Real-time data management

12. Future Enhancements

- Add notifications and reminders
- Implement dark mode
- Add task categories and labels
- Real-time collaboration using WebSockets
- Mobile application support

13. Conclusion

The Full Stack Task Manager project demonstrates practical implementation of modern web development concepts. It combines frontend, backend, and database technologies into a complete working application. The project is useful for learning CRUD operations, authentication, API integration, and deployment.

14. Technology Summary Table

Module	Technology
Frontend	React.js
Backend	Node.js + Express.js
Database	MongoDB
Authentication	JWT
Deployment	Vercel + Render